HAROL MAXILAN

Data Science Undergraduate

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PROFILE

Data Science undergraduate at Sabaragamuwa University of Sri Lanka focused on data engineering and AI/ML. Built a Medallion data pipeline over 2.3M+ transactions, a fine-tuned BERT emotion classifier (0.88 Macro F1), and a multi-agent RAG system on LangGraph. Seeking a Data Engineering, AI/ML, or Data Science internship.

EDUCATION

BSc (Hons) in Data Science — Sabaragamuwa University of Sri Lanka, Faculty of Computing **2024 – 2028 (Expected)**

Coursework: Database Systems, Data Warehousing, Data Science, Statistics, Machine Learning, Artificial Intelligence

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, PHP

Data Engineering: ETL/ELT, Medallion Architecture, Star/Snowflake Schema, Data Quality Auditing, Geospatial Features, Batch Processing

ML, DL & NLP: Scikit-learn, PyTorch, MLflow, SMOTE, Hugging Face Transformers, BERT, DistilBERT, TF-IDF, Hyperparameter Tuning

Generative AI: LangChain, LangGraph, Multi-Agent Systems, RAG, Prompt Engineering, FAISS, ChromaDB, Pydantic, Google Generative AI

Computer Vision: OpenCV, MediaPipe, CNNs, Real-Time Inference

Databases & Analysis: MySQL, PostgreSQL, MariaDB, Pandas, NumPy, Matplotlib, Plotly, EDA

Cloud & Tools: FastAPI, Streamlit, Docker, Git, Linux, CI/CD, REST APIs, Pytest

PROJECTS

CareerCompass AI — Multi-Agent Career Coaching System | *Group project* **2026**

Tech: *Python, LangGraph, LangChain, RAG, FAISS, FastAPI, Pydantic | github.com/sankajithdjinasena/CareerCompass_AI*

- Co-developed a five-agent coaching system on LangGraph covering CV analysis, skill-gap assessment, mock interviews, job scraping, and roadmap generation.
- Implemented a FAISS and Tavily RAG pipeline that matches candidate profiles to live job postings, with Pydantic data validation.
- Introduced self-correcting JSON validation loops, reaching 100% schema compliance across 50+ simulated interview sessions.

F1 Super Intelligence — Analytics & Precomputation Pipeline | *Personal project* **2025**

Tech: *Python, Pandas, Plotly, Pytest, Batch Processing | github.com/Max-i26/F1-Super-Intelligence-System*

- Constructed a batch ETL pipeline precomputing 1.2M+ telemetry points across 24 Grand Prix rounds for sub-second dashboards.
- Wrote a 56-test Pytest audit suite validating sensor completeness, race strategies, and lap-timing anomalies.
- Modeled pace degradation and tire telemetry in interactive Plotly dashboards, cutting query latency from 4.2 s to under 350 ms.

NLP-Based Emotion Detection Pipeline | *Personal project* **2025**

Tech: *Python, BERT, DistilBERT, PyTorch, Hugging Face, MLflow, SMOTE | github.com/Max-i26/NLP-Emotion-Detection*

- Benchmarked a TF-IDF baseline (0.76 F1) against fine-tuned BERT and DistilBERT on 20,000+ text samples, tracking every run in MLflow.
- Countered severe class imbalance with SMOTE, class-weighted loss, and label smoothing, lifting minority-class recall by 24%.
- Reached 0.88 Macro F1 and 91.2% Weighted F1 on held-out data, a 15.8% improvement over the baseline.

Friday — No-Code AutoML & AI Tutor Platform | *Personal project* **2025**

Tech: *Python, Scikit-learn, Docker, FastAPI, Chart.js, Gemini API | github.com/Max-i26/Friday*

- Containerized a no-code AutoML web app (Docker, FastAPI) that ingests CSV/Excel files up to 150 MB and profiles them instantly.
- Trained Linear/Logistic, Polynomial, Decision Tree, and Random Forest models in parallel with auto-detected task type and a leaderboard.
- Integrated the Gemini API as an AI tutor that explains datasets and audits models for overfitting and data leakage.

Image-Based Emotion Detection | *Personal project* **2025**

Tech: *Python, OpenCV, CNN, Streamlit | github.com/Max-i26/emotion_detection*

- Trained a deep CNN on 35,000+ FER-2013 images, reaching 87.4% validation accuracy across 7 emotion classes.
- Built real-time face detection with OpenCV Haar Cascades and a Streamlit app for batch and live-webcam analysis at under 25 ms latency.

Additional Projects

- **Syndicate DataStorm (2025, team):** Designed a Python Bronze–Silver–Gold pipeline that cleaned 2.3M+ transactions, quarantined ~66K invalid records, and added competition and OpenStreetMap POI features to predict outlet potential.
- **NexLab (2025 – 2026, Team Leader, CIPHER 2.0 Finalist):** Led a PHP/MySQL lab-allocation platform for 500+ requests with priority scoring and zero double-bookings.
- **BEE (2025):** Created a MediaPipe/OpenCV gesture interface running at 30+ FPS with 96% accuracy across 7 gestures.

AWARDS & ACHIEVEMENTS

- **Nexora Ideathon — 3rd Place** | Sabaragamuwa University of Sri Lanka (2025)
- **Hackathon Finalist** | CIPHER 2.0 (IIT), J'PuraXtreme 3.0 (USJ), CodeRally 7.0 (IIT, Intermediate Tier), Code Arena (KDU) (2025)

LEADERSHIP & VOLUNTEER EXPERIENCE

- **AWS Logistics and Operations Lead**, Sabaragamuwa University (2026 – Present) | **Zone C University Ambassador, IEEE INSL** (2025)
- **Club Cricket Team Captain**, VSC, Vavuniya (2025) | **Chess Competition Mentor**, Clash of Pawns, AUSDAV (2025)

REFERENCES

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